

Vision Document: eRegistrar

Course: CS425, Software Engineering

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1. Introduction

Course registration in the Computer Science department is still run on spreadsheets and email. The registrar builds each term's schedule by hand, faculty send in their teaching preferences by email, and students register on paper forms that staff later key into the system.

That worked fine when the department was small. It now runs **four entries a year** with 100 to 130 students per entry, and offers 8 or 9 electives in a busy block. Faculty each have one or two specializations, a set of courses they can teach, and preferences about which blocks they're available in. Some 500-level courses have 400-level prerequisites, so the 400-level offerings need to land in a student's earlier blocks. Keeping all of this straight in a spreadsheet is slow, and prerequisite or capacity problems usually turn up only after students discover them, once registration is already open.

eRegistrar is a web application that builds the term schedule with faculty assigned to each section, and lets students register online with the prerequisite and capacity rules enforced automatically.

2. Positioning

2.1 Problem Statement

The problem of	building the term schedule, assigning qualified faculty, and registering students for classes
Affects	the registrar, faculty, and students
The impact of which is	scheduling takes weeks of manual work, prerequisite and capacity violations are found late, and every schedule change forces a manual re-check
A successful solution would be	one system holding courses, faculty profiles and the schedule in a single database, applying the rules automatically, with web access for administrators, faculty and students

2.2 Product Position Statement

For	the registrar, faculty and students of the Computer Science department
Who	need a conflict-free term schedule and online registration against it
The eRegistrar	is a web-based course scheduling and registration system
That	generates the schedule from faculty qualifications and availability, and enforces prerequisites and capacity at the moment of registration
Unlike	spreadsheets plus email plus manual data entry
Our product	keeps one authoritative schedule, so a change made once is reflected everywhere and invalid registrations are rejected immediately

3. Stakeholders

Stakeholder	Role in the system
Registrar (Administrator)	Maintains courses and programs, generates and publishes the schedule, assigns faculty.
Faculty	Maintains their own profile: specializations, teachable courses, block availability. Views their assigned sections.
Student	Views the published schedule, registers for and drops courses.
Department Chair	Approves the published schedule. Project sponsor.
IT Administrator	Deploys and operates the system, manages accounts.

User environment. Two to three registrar staff, about 25 faculty, and up to 500 students on campus. All access is through a browser on desktop or phone, with no software installed on user machines. Registration load is heaviest in the first hour after the registration window opens.

4. Product Overview

4.1 Needs and Features: Problem / Need / Feature Table

#	Problem	Need	Priority	Feature
1	Faculty preferences arrive by email and are applied inconsistently	Faculty must own their qualifications and availability	High	F1: Faculty Profile Management
2	Course and prerequisite data live in several spreadsheets that disagree	One authoritative catalog	High	F2: Course & Program Catalog
3	Building a term schedule takes weeks of manual work	Generate the schedule automatically from the rules	High	F3: Schedule Generation
4	Manual assignment double-books faculty or assigns courses they can't teach	Detect assignment conflicts	High	F4: Faculty Assignment & Conflict Detection
5	Students register on paper forms that staff re-key	Students register themselves with an immediate answer	High	F5: Online Student Registration
6	Prerequisite and capacity violations are found after registration closes	Invalid registrations rejected as they are attempted	High	F6: Registration Rule Enforcement
7	Nobody can see live enrolment	Role-specific views of the schedule and enrolment	Medium	F7: Schedule & Enrolment Views
8	Anyone with the spreadsheet can change anything	Access must match the user's role	High	F8: Authentication & Role-Based Access

4.2 Assumptions and Dependencies

1. The academic calendar of blocks and entries is fixed by the university.
2. Course and program definitions are maintained inside eRegistrar by the registrar.
3. Every user has a university account; authentication is delegated to the campus identity provider.
4. The deployment target is a Java/Spring Boot environment with a relational database.

4.3 Alternatives

- **Status quo (spreadsheets and email).** Free and familiar, but it has no rule enforcement, no concurrency control, and won't scale to four entries a year.
- **Commercial SIS registration modules.** Comprehensive, but priced and scoped for a whole institution, and they don't model this department's block/entry structure without heavy customization.
- **Build in-house.** This is what we chose. The scheduling rules are specific to this department and are the hard part of the problem, so a small focused application ends up cheaper than customizing a large product.

5. Other Product Requirements

Category	Requirement
Platform	Java 11+, Spring Boot, relational database, browser client
Performance	Pages respond within 2 seconds; 150 concurrent users during registration
Concurrency	Section capacity holds under simultaneous registration
Security	Role-based authorization on every operation, TLS everywhere, no passwords stored by the application
Usability	Student pages usable on a phone; staff productive after one walkthrough
Compliance	Student records handled in accordance with FERPA